



CSCE 633-700: Machine Learning

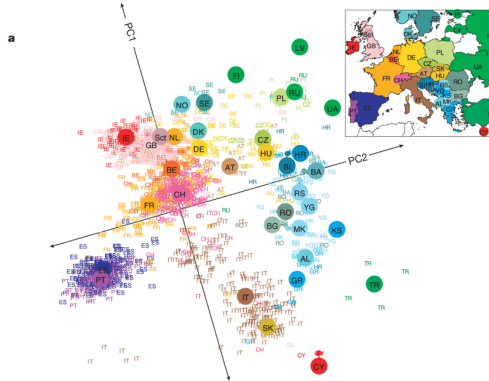
Clustering

Overview

- Clustering overview
- Partitional clustering
 - K-means clustering
 - Gaussian Mixture Models (GMM)
- Hierarchical clustering

Clustering

(1) Understanding: Finding patterns/structure/sub-populations in data



(2) Summarization: Reducing the size of large datasets

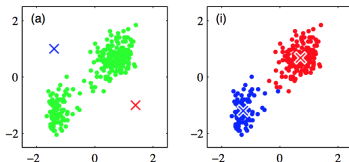
Overview

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- **Input:** Data $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$
- **Output:** Clusters $\mu_1, \dots, \mu_K$
- **Decision:** Cluster membership, the cluster id assigned to sample $\mathbf{x}_n$, i.e. $A(\mathbf{x}_n) \in \{1, \dots, K\}$
- **Evaluation metric:** Distortion measure

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \mu_k\|_2^2, \text{ where } r_{nk} = 1 \text{ if } A(\mathbf{x}_n) = k, 0 \text{ otherwise}$$

- **Intuition:** Data points assigned to cluster k should be close to centroid μ_k



K-means Clustering

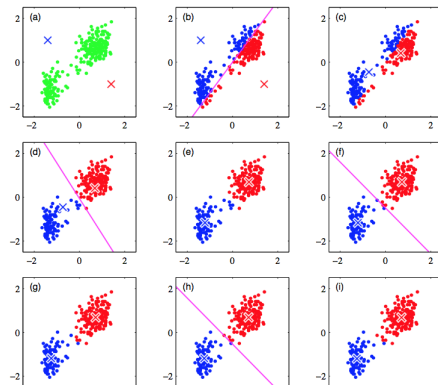
Evaluation metric:
$$\min_{r_{nk}} J = \min_{r_{nk}} \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|_2^2$$

Optimization:

- **Step 0:** Initialize $\boldsymbol{\mu}_k$ to some values
- **Step 1:** Assume the current value of $\boldsymbol{\mu}_k$ fixed, minimize J over r_{nk} , which leads to the following cluster assignment rule $r_{nk} = \begin{cases} 1, & \text{if } k = \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|_2^2 \\ 0, & \text{otherwise} \end{cases}$
- **Step 2:** Assume the current value of r_{nk} fixed, minimize J over $\boldsymbol{\mu}_k$, which leads to the following rule to update the prototypes of the clusters $\boldsymbol{\mu}_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}$
- **Step 3:** Determine whether to stop or return to Step 1

K-means Clustering

Example



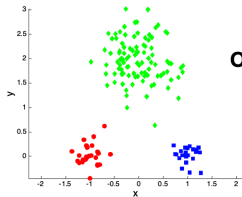
K-means Clustering

Remarks

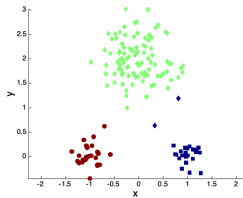
- The centroid μ_k is the means of data points assigned to the cluster k , hence the name K-means clustering.
- The procedure terminates after a finite number of steps, as the procedure reduces J in both Step 1 and Step 2
- There is no guarantee the procedure terminates at the global optimum of J . In most cases, the algorithm stops at a **local optimum**, which depends on the initial values in Step 0 $\rightarrow$ **random restarts** to improve chances of getting closer to global optima

K-means Clustering

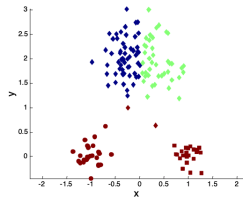
Initialization of K-Means is important



Original Points



Optimal Clustering



Sub-optimal Clustering

K-means Clustering

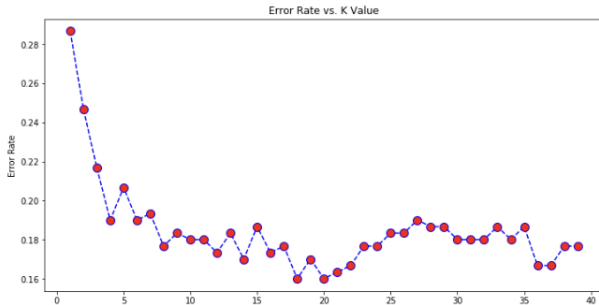
Solutions to Initial Centroids Problem

- Multiple random initializations
- Start with hierarchical clustering to determine initial centroids
- Select more than K initial centroids and then select among these initial centroids

K-means Clustering

How to know when to stop - Elbow Method

- Plot the error (i.e., distance of each sample to the corresponding centroid) against the number of clusters
- Stop when the decrease in error becomes almost flat



K-means Clustering

Application: vector quantization

- We can replace our data points with the centroids μ_k from the clusters they are assigned to $\rightarrow$ **vector quantization**
- We have compressed the data points into
 - a codebook of all the centroids $\{\mu_1, \dots, \mu_K\}$
 - a list of indices to the codebook for the data points (created based on r_{nk})
- This compression is obviously lossy as certain information will be lost if we use a very small K

K-means Clustering

Limitations of K-Means

- Problems when clusters are of differing size, density, or non-spherical shapes (for Euclidean distances)
- Sensitive to outliers
- Number of clusters is difficult to determine